PAPER_1612 — U-235 Binding Energy per A = U-235 BE/A = $F \cdot K^5 + \beta + F \cdot \beta + 3 \approx 7.588$ MeV

Author: Daniel T. Murphy **Framework:** UQFF (Unified Quantum Field Framework) — Star-Magic v5.27+ **Tier:** Nuclear **Date:** June 18, 2026 **Status:** CLOSED — 0.043% residual

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Observation

PAPER_1209II_S666 (Nuclear Unified Proof Set) gives the closed-form expression:

```
U-235 BE/A = F·K⁵ + β + F·β + 3 ≈ 7.588 MeV
```

Residual to AME2020: **0.043%**.

UQFF Closed Identity

```
U-235 BE/A = F·K⁵ + β + F·β + 3 ≈ 7.588 MeV    0.043%
```

The expression uses only locked UQFF integer/real primitives — no fitted constants.

NOT REPLACEMENT

Experimental nuclear measures this constant directly. UQFF supplies a structural derivation from the integer primitives.

Reference

- Source: PAPER_1209II_S666
- Related: PAPER_1494-1605 (other session-2026-06-18 closures)
- Calculator dispatch: `calculate_paradox({"paradox": "nuclear_u235_be_a_7_591"})`
